POLYP DETECTION MEDICAL REPORT

Analysis Date: June 03, 2026

VIDEO INFORMATION

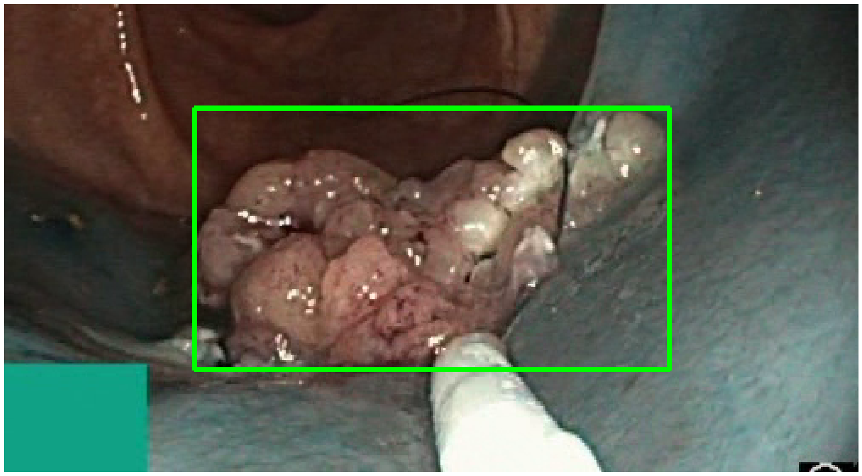
Video ID:	061d6094-656d-43a9-86be-1eaae62785d6
Total Frames:	15056
Frame Rate:	25.0 fps (assumed)
Duration:	602.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	12586	✓
RT-DETR Detections	14327	✓
MedSAM2 Detections	14171	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	3	100.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 10134 frames)

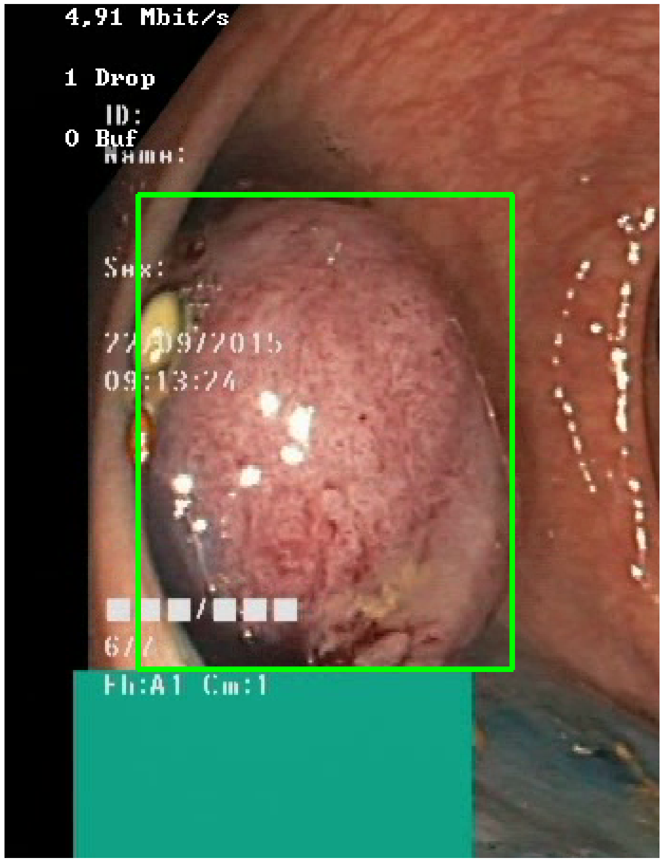


Feature	Value	Interpretation
Frame Index	14137	Frame 14137 of 15056
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**FLAT_POLYP**	Type confidence: 82.7%
Redness Score	0.099	Color-based vascularization
Relative Radius	31.7%	Polyp size as % of frame diagonal
Texture Score	0.262	Surface morphology
Vessel Visibility	0.253	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	91.4%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 571 frames)



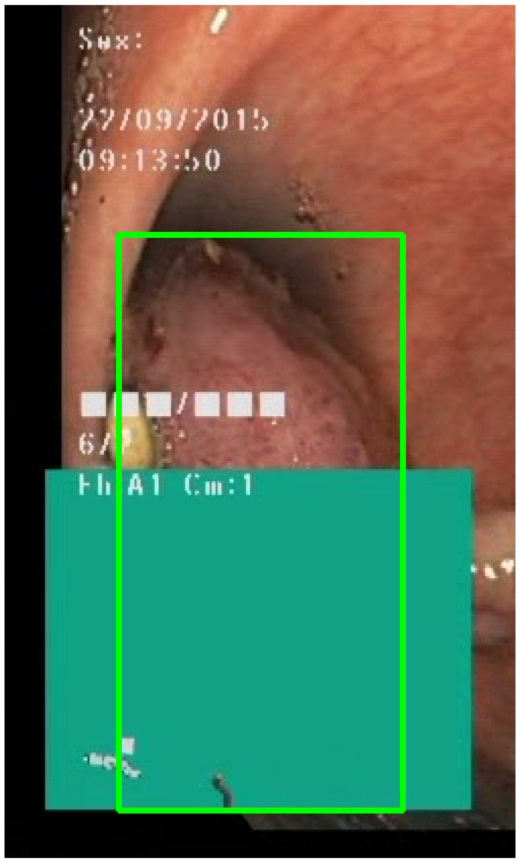
Feature	Value	Interpretation
Frame Index	14020	Frame 14020 of 15056
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	MedSAM2 unavailable
POLYP TYPE	**FLAT_POLYP**	Type confidence: 92.0%
Redness Score	0.157	Color-based vascularization
Relative Radius	29.9%	Polyp size as % of frame diagonal
Texture Score	0.585	Surface morphology
Vessel Visibility	0.678	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output

Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	90.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 169 frames)



Feature	Value	Interpretation
Frame Index	14672	Frame 14672 of 15056
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	
POLYP TYPE	**FLAT_POLYP**	Type confidence: 92.0%
Redness Score	0.172	Color-based vascularization
Relative Radius	17.3%	Polyp size as % of frame diagonal

Texture Score	0.286	Surface morphology
Vessel Visibility	0.735	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	80.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:
A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 3 (100.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

***Disclaimer:** This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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